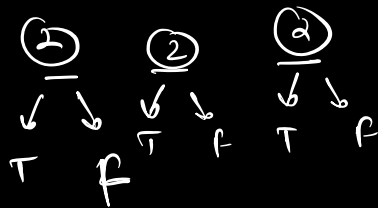


Q1. Given 3 T/F questions, every question has to answered
How many ways can we ans all the questions?

$\begin{matrix} T T T \\ T T F \\ T F T \\ T F F \\ F F F \\ F T T \\ F T F \\ F F T \end{matrix}$

8



$$\Rightarrow 2 \times 2 \times 2 = 8$$

$\downarrow$
 2^3

examples

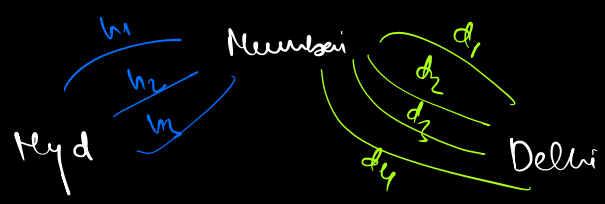
1) Given 10 girls and 7 boys, How many diff.

$\begin{matrix} \text{pairs} \\ \swarrow \searrow \\ \text{girl} \quad \text{boy} \end{matrix}$

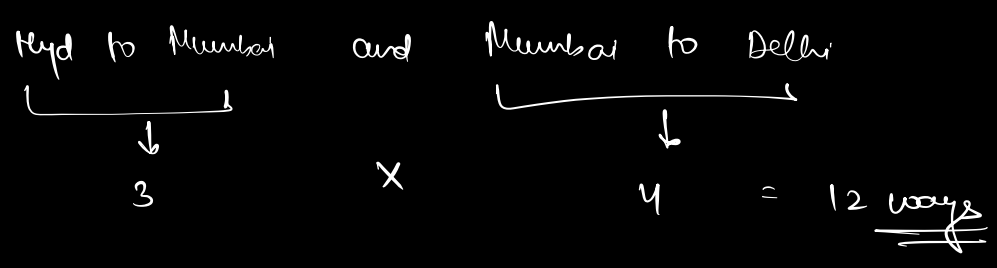
girl & boy

$$\left(\begin{matrix} 10 \text{ girls} \\ \text{to} \end{matrix} \right) \times \left(\begin{matrix} 7 \text{ boys} \\ \text{to} \end{matrix} \right) \Rightarrow 70 \text{ ways/pairs}$$

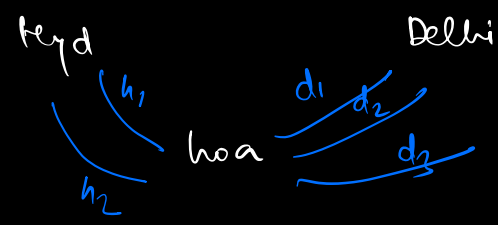
ex = 2



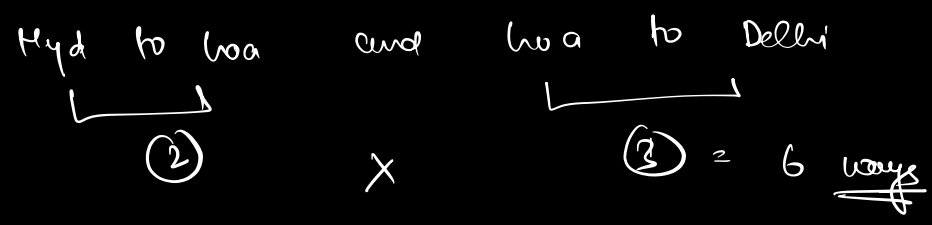
Hyd to Delhi via Mumbai



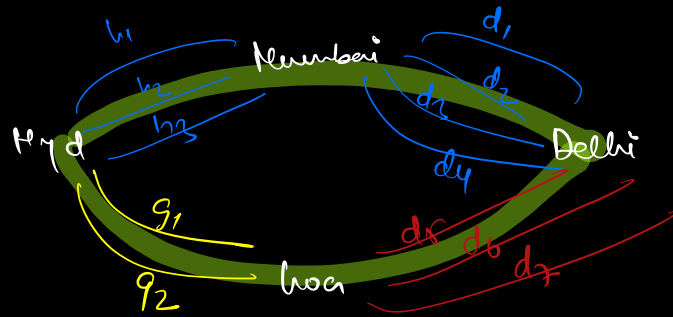
ex = 3 $\Rightarrow$



Hyd to Delhi via Goa



ex $\Rightarrow 4 \Rightarrow$



Hyd to Delhi via Mumbai $\rightarrow 12$

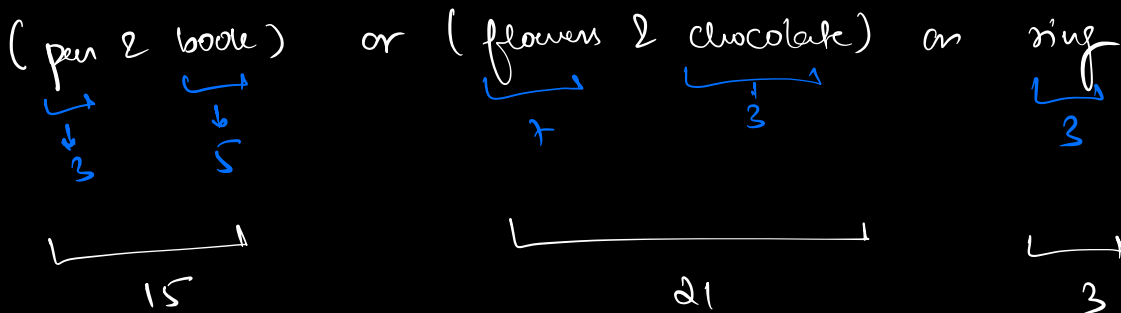
or,

$\uparrow$

Hyd to Delhi via Goa $\rightarrow 6$

18 ways

If you need to gift-



$$\text{Total ways} = 15 + 21 + 3 = 39$$

⇒ Permutations :- arrangement of objects

$$[(i, j) \neq (j, i) \rightarrow \text{arrangement}]$$

Q1, Given 3 distinct characters, how many ways can we arrange them?

S = "a c d"

a c d
a d c
d c a
d a c
c d a
c a d

6 ways

$$\begin{array}{ccc} \underline{3} & \underline{2} & \underline{1} \\ \downarrow & \downarrow & \\ \{a\} & \{c, d\} & \\ \{c\} & \{a, d\} & \\ \{d\} & \{a, c\} & \end{array} \Rightarrow 3 \times 2 \times 1 = \textcircled{6}$$

⇒ if N distinct elements are given, how many ways can we arrange them?

$$\underline{N} \quad \underline{N-1} \quad \underline{N-2} \quad \underline{N-3} \quad \dots \quad \underline{1}$$

$$N \times N-1 \times N-2 \times N-3 \dots \times 1 =$$

$$\textcircled{N!}$$

⇒ Given 5 distinct characters, how many ways can be arranged 2 characters.

⇒ a b c d e

$\begin{array}{c} 5 \\ \hline \downarrow \end{array}$
 $\begin{array}{c} 4 \\ \hline \end{array}$
 $\Rightarrow 5 \times 4 = 20 \text{ ways}$

$\left\{ \begin{array}{l} \textcircled{a} \\ \textcircled{b} \\ c \\ d \\ e \end{array} \right\}$
 $\begin{array}{l} (bcde) \\ (acde) \\ (abde) \\ (abce) \\ (abcd) \end{array}$

$\left[\begin{array}{ccccc} ab & ba & ca & da & ea \\ ac & bc & cb & db & eb \\ ad & bd & cd & dc & ec \\ ae & be & ce & de & ed \end{array} \right] \rightarrow 20 \text{ ways}$

arrange 2 chars, out of N

$$\underline{N} \quad \underline{N-1} \quad \Rightarrow \quad N(N-1) \text{ ways}$$

arrange 3 chars, out of N

$$\underline{N} \quad \underline{N-1} \quad \underline{N-2} \quad \Rightarrow \quad N(N-1)(N-2) \text{ ways}$$

1) N distinct elements, arrange r element —

$$N(N-1)(N-2)(N-3) \dots (N-(r-1))$$

$$\Rightarrow \frac{N(N-1)(N-2)(N-3) \dots (N-r+1) \times (N-r) \times (N-r-1) \times \dots \times 1}{(N-r) \times (N-r-1) \times \dots \times 1}$$

$$\Rightarrow \boxed{\frac{N!}{(N-r)!} \Rightarrow {}^N P_r}$$

$\Rightarrow$ **Combinations** $\Rightarrow$ **selection** of objects

$$(i, j) = (j, i) \rightarrow \text{arrangement doesn't matter}$$

$B_1 \ B_2 \ B_3 \ B_4 \] \rightarrow$ out of 4 boys, select 3 boys

$$B_1 \ B_2 \ B_3 \} = B_2 \ B_3 \ B_1 = B_3 \ B_1 \ B_2$$

$$B_1 \ B_2 \ B_4$$

$$B_1 \ B_3 \ B_4$$

$$B_2 \ B_3 \ B_4$$

$\Rightarrow$ 4 ways to select 3 boys out of 4

2) No. of ways we can arrange 3 boys from 4 boys

$$\text{arrangement} = {}^4P_3 = \frac{4!}{(4-3)!} = 24 \text{ ways}$$

1) find no. of ways we can select 3 boys

<u>B₁ B₂ B₃</u>	<u>B₁ B₂ B₄</u>	<u>B₁ B₃ B₄</u>	<u>B₂ B₃ B₄</u>
B ₁ B ₂ B ₃	B ₁ B ₂ B ₄		
B ₁ B ₃ B ₂	B ₁ B ₄ B ₂		
B ₂ B ₁ B ₃	B ₂ B ₄ B ₁		
B ₂ B ₃ B ₁	B ₂ B ₁ B ₄		
B ₃ B ₁ B ₂	B ₄ B ₁ B ₂		
B ₃ B ₂ B ₁	B ₄ B ₂ B ₁		

↳ 6 ways ↳ 6 ways

total no. of combis (groups) possible arrangement of all items of each group.

total ways $\Rightarrow \underline{4 \times 6 = 24 \text{ ways}}$

no. of possible combis / groups $\Rightarrow x$

size of each group $\Rightarrow r \Rightarrow r!$ ← no. of arrangements

$$\text{total no. of arrangements} \Rightarrow x \times r! \Rightarrow {}^N P_r$$

$$\Rightarrow x \times r! = \frac{N!}{(N-r)!}$$

$$\Rightarrow x = \frac{N!}{(N-r)! r!}$$

So, total no. of combinations of r items

$$\text{out of } N \text{ items} \Rightarrow {}^N C_r = \frac{N!}{(N-r)! r!}$$

∴ Properties of ${}^N C_r$:

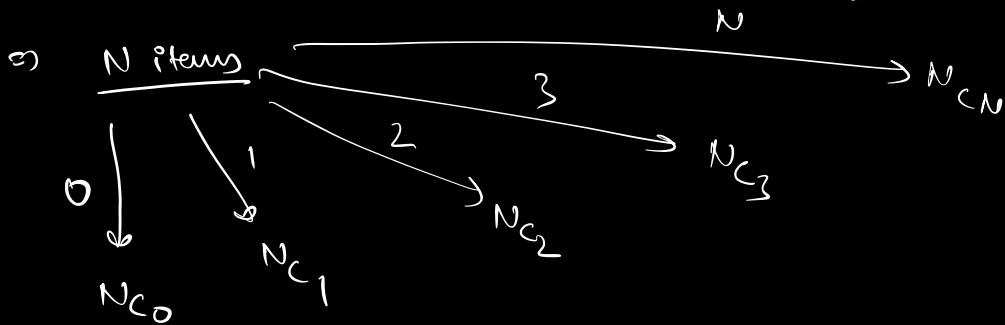
$${}^N C_1 \Rightarrow \frac{N!}{(N-1)! 1!} = \frac{N!}{(N-1)!} = \frac{N \times \cancel{(N-1)!}}{\cancel{(N-1)!}} = N$$

$${}^N P_1 \Rightarrow \frac{N!}{(N-1)!} = N$$

$$N_{C0} \Rightarrow \frac{N!}{(N-0)! \cdot 0!} = \frac{N!}{N!} = 1$$

$$\{0! = 1\}$$

total possible ways of selecting any no. of items



$$N_{C0} + N_{C1} + N_{C2} + N_{C3} + \dots + N_{CN} = 2^N$$

$a_1 \ a_2 \ a_3$

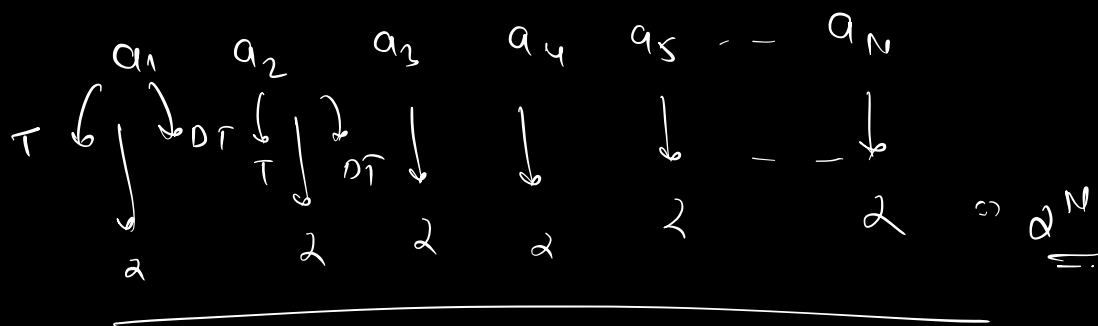
$$\{\} \rightarrow N_{C0} \quad \text{①}$$

$$\begin{matrix} \{a_1\} \\ \{a_2\} \\ \{a_3\} \end{matrix} \rightarrow N_{C1} \quad \text{②}$$

$$\begin{matrix} \{a_1, a_2\} \\ \{a_2, a_3\} \\ \{a_1, a_3\} \end{matrix} \rightarrow N_{C2} \quad \text{③}$$

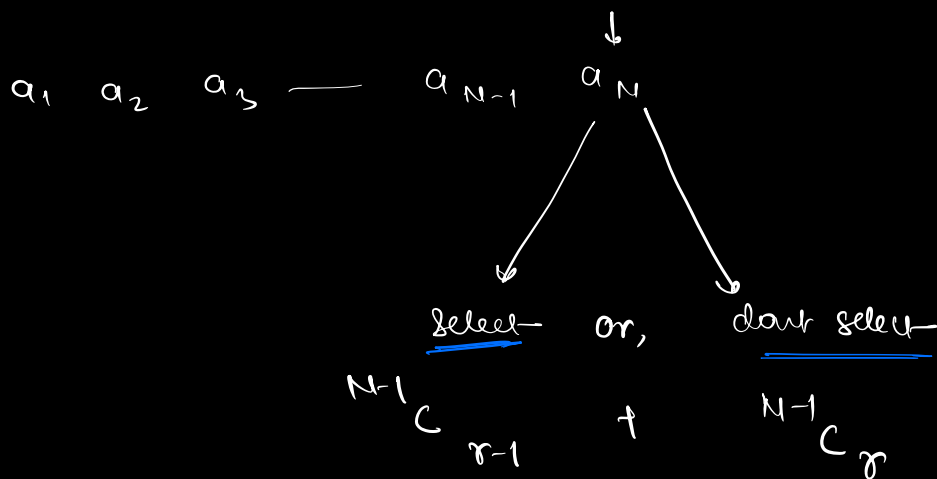
$$\{a_1, a_2, a_3\} \rightarrow N_{C3} \quad \text{④}$$

$$\text{total} = 1 + 3 + 3 + 1 = 8 \Rightarrow \underline{\underline{2^3}}$$



$${}^N C_R = \frac{N!}{(N-R)! R!}$$

⇒ given N , how many ways we can select r items:



$${}^N C_r = {}^{N-1} C_{r-1} + {}^{N-1} C_r$$

$$\frac{(N-1)!}{(N-1-(r-1))! (r-1)!} + \frac{(N-1)!}{(N-1-r)! r!}$$

$$= \frac{(N-1)!}{(N-r)!(r-1)!} + \frac{(N-1)!}{(N-r-1)!r!}$$

$$\left\{ \begin{array}{l} r! \leq r \times (r-1)! \\ (N-r)! \\ = (N-r) \times (N-r-1)! \end{array} \right.$$

$$= \frac{(N-1)!}{(N-r)(N-r-1)!(r-1)!} + \frac{(N-1)!}{(N-r-1)!(r)(r-1)!}$$

$$= \frac{(N-1)!}{(r-1)!(N-r-1)!} \left[\frac{1}{N-r} + \frac{1}{r} \right]$$

$$= \frac{(N-1)!}{(r-1)!(N-r-1)!} \left[\frac{r + N-r}{r(N-r)} \right]$$

$$= \frac{N(N-1)!}{r(r-1)!(N-r)(N-r-1)!}$$

$$= \frac{N!}{r!(N-r)!} \Rightarrow N C_R$$

If 5 girls, you need to select 2 girls, find no. of ways:

$$G_1 \quad G_2 \quad G_3 \quad G_4 \quad G_5 \rightarrow G_{C_2} \Rightarrow \frac{5!}{(5-2)! 2!}$$

$$G_1 \quad G_2 \rightarrow G_3 \quad G_4 \quad G_5 \quad G_3 \quad G_4 \rightarrow G_1 \quad G_2 \quad G_5$$

$$G_1 \quad G_3 \rightarrow G_2 \quad G_4 \quad G_5 \quad G_2 \quad G_5 \rightarrow G_1 \quad G_2 \quad G_4$$

$$G_1 \quad G_4 \rightarrow G_2 \quad G_3 \quad G_5$$

$$G_1 \quad G_5 \rightarrow G_2 \quad G_3 \quad G_4 \quad G_4 \quad G_5 \rightarrow G_1 \quad G_2 \quad G_3$$

$$G_2 \quad G_3 \rightarrow G_1 \quad G_4 \quad G_5$$

$$G_2 \quad G_4 \rightarrow G_1 \quad G_3 \quad G_5$$

$$G_2 \quad G_5 \rightarrow G_1 \quad G_3 \quad G_4$$

$$= \frac{5 \times 4 \times 3 \times 2}{2 \times 2}$$

$$= 10$$

$$G_{C_2} = G_{C_3}$$

→ N girls, selection of r $\Rightarrow N_{C_r}$

→ N girls, rejection of N-r $\Rightarrow N_{C_{N-r}}$

$$N_{C_r} = N_{C_{N-r}}$$